2017 E.C. National EUEE Mathematics

EUEE Entrance Examination Archive â¢ Mathematics

Federal Democratic Republic of Ethiopia - Ministry of Education

Curriculum Development and Implementation Directorate

Official Student Textbook for Secondary Education - Grades 9 to 12

National Syllabus Aligned with University Entrance Examination Standards

Verified with high-resolution digital text extraction and OCR diagram references.

TABLE OF CONTENTS & CURRICULUM SYLLABUS

Unit 1: Foundational Principles and Core Concepts ..... Pages 1 - 35

Unit 2: Analytical Methods, Formulas, and Equations ..... Pages 36 - 75

Unit 3: Scientific Investigation & Applied Problem Solving ..... Pages 76 - 115

Unit 4: Advanced Theorems, Kinetics, and Models ..... Pages 116 - 155

Unit 5: National Entrance Examination Review & Solutions ..... Pages 156 - 200

Each unit includes theory, worked examples, exercises, and EUEE exam alignments.

Section 3.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

Section 4.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

2017 E.C. National EUEE Mathematics - Unit 2

Section 5.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

2017 E.C. National EUEE Mathematics - Unit 2

Section 6.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

Section 7.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

Section 8.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

Section 9.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

2017 E.C. National EUEE Mathematics - Unit 4

Section 10.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

2017 E.C. National EUEE Mathematics - Unit 5

Section 11.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

2017 E.C. National EUEE Mathematics - Unit 5

Section 12.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

2017 E.C. National EUEE Mathematics - Unit 5

Section 13.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

Section 14.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

Section 15.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

Section 16.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

2017 E.C. National EUEE Mathematics - Unit 5

Section 17.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

2017 E.C. National EUEE Mathematics - Unit 5

Section 18.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

Section 19.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

Section 20.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

2017 E.C. National EUEE Mathematics - Unit 5

Section 21.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

Section 22.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

2017 E.C. National EUEE Mathematics - Unit 5

Section 23.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.

2017 E.C. National EUEE Mathematics - Unit 5

Section 24.1: Core Theoretical Principles & Systematic Derivations

1. Theoretical Foundation:

According to the Ethiopian national curriculum standards, students must master the mathematical relationships, scientific observations, and key definitions.

2. Worked Example & Problem Solving Protocol:

Given the standard conditions, apply the fundamental formula to solve step-by-step. Verify the units, dimensional consistency, and numerical precision of the result.

3. Review Questions & University Entrance Examination Connection:

Solve practice exercises 1 through 5 at the end of this curriculum section.